IMEN574 / IDSC574 · PROGRAMMING FOR AX

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Visual Representations

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Where We Are

01 **Programming with AI**

Lectures 2–4

02 **Python and Classical Machine Learning**

Lectures 5–13

03 **Deep Learning and Foundation Models**

Lectures 14–22

Outline: Visual Representations

01 **Encoding Images**

From pixels to tokens

02 **Learning Representations**

From pairs to features

03 **Transferring Features**

From features to tasks

01 · Encoding Images

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From features to tasks

Why Representations

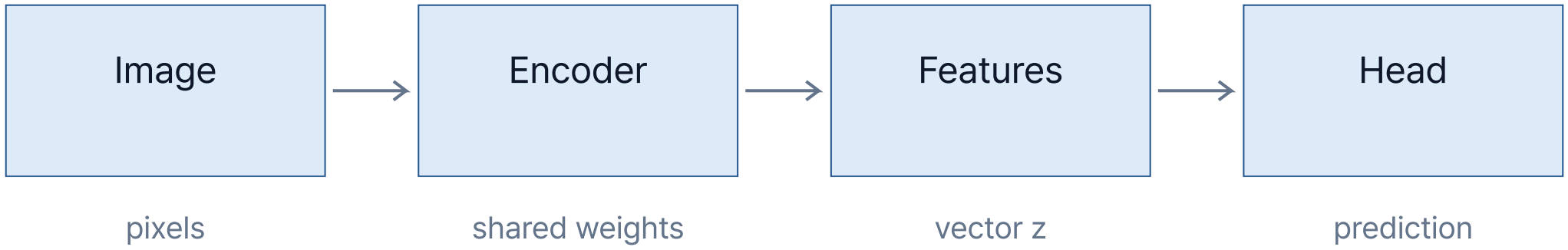
Pixel distance changes when the object barely changes

Image A	0	0	1	0
Shifted A	0	1	0	0
Background	0	0	0	0

A one-pixel shift changes the array

Representation and Readout

The encoder supplies features to a task-specific head



A new head can reuse the same features

Global and Spatial Features

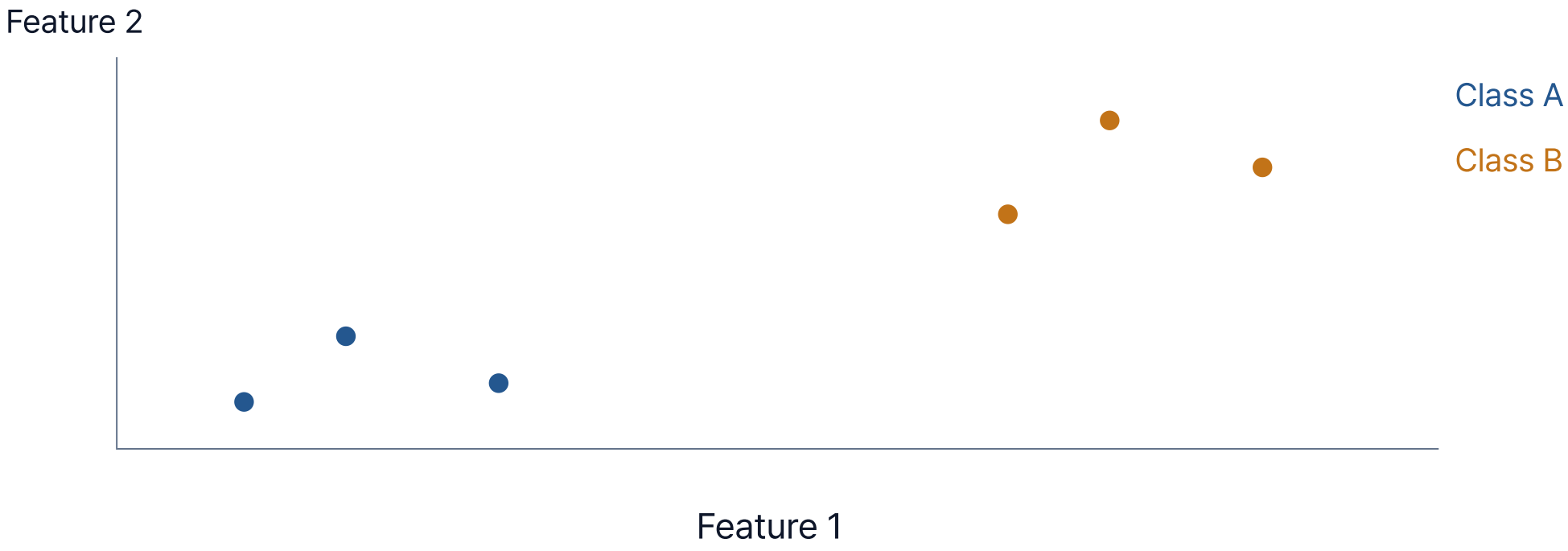
Pooling trades location for a compact summary

Patch 1	1	0	0	1
Patch 2	0	1	1	0

Pooling removes the identity of each patch

Supervised Visual Learning

Labels shape which differences matter



Nearby embeddings share the training label

Local Receptive Fields

A convolution applies one local rule across an image

top	0	1	0
middle	1	4	1
bottom	0	1	0

The same kernel visits each local neighborhood

Convolution Output Size

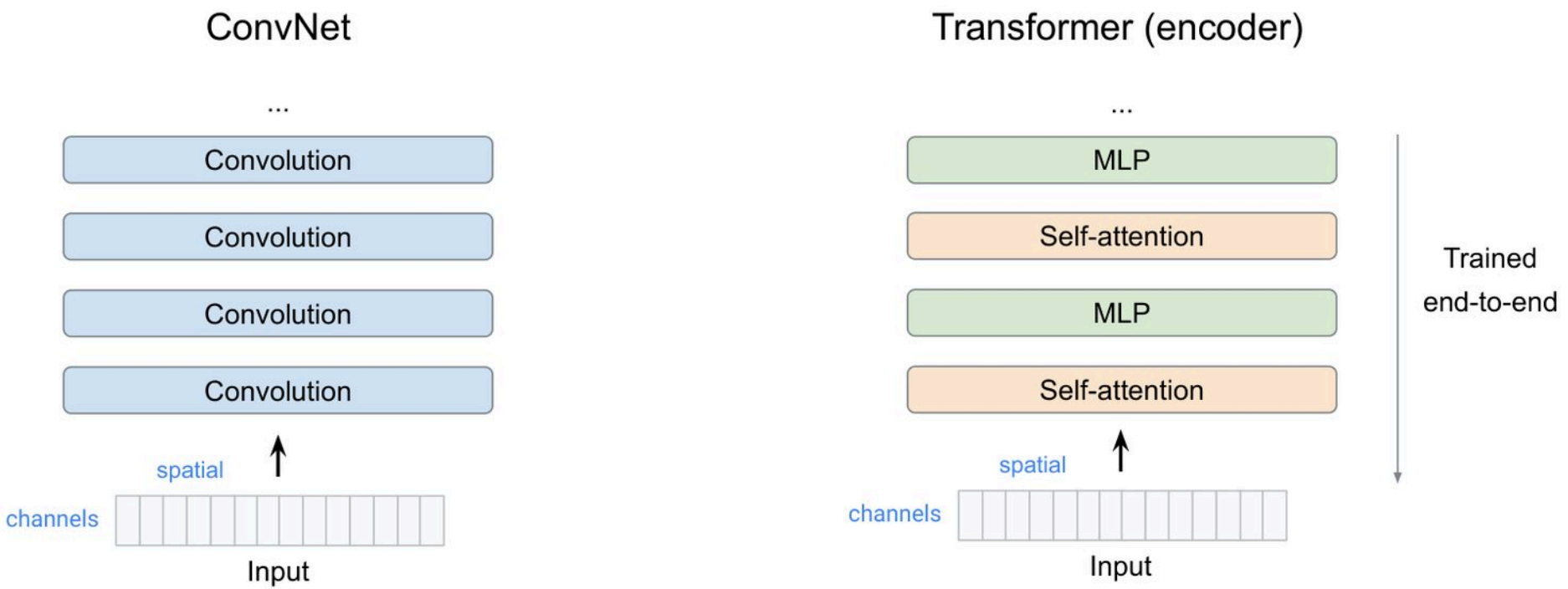
Stride and padding determine the spatial grid

$$H_{out} = \text{floor}((H + 2P - K) / S) + 1$$



Convolution and Attention

Local filters and global token mixing impose different structures



Token mixing changes which locations can interact

https://arxiv.org/abs/2010.11929

Patch Embeddings

An image becomes a sequence of patch vectors

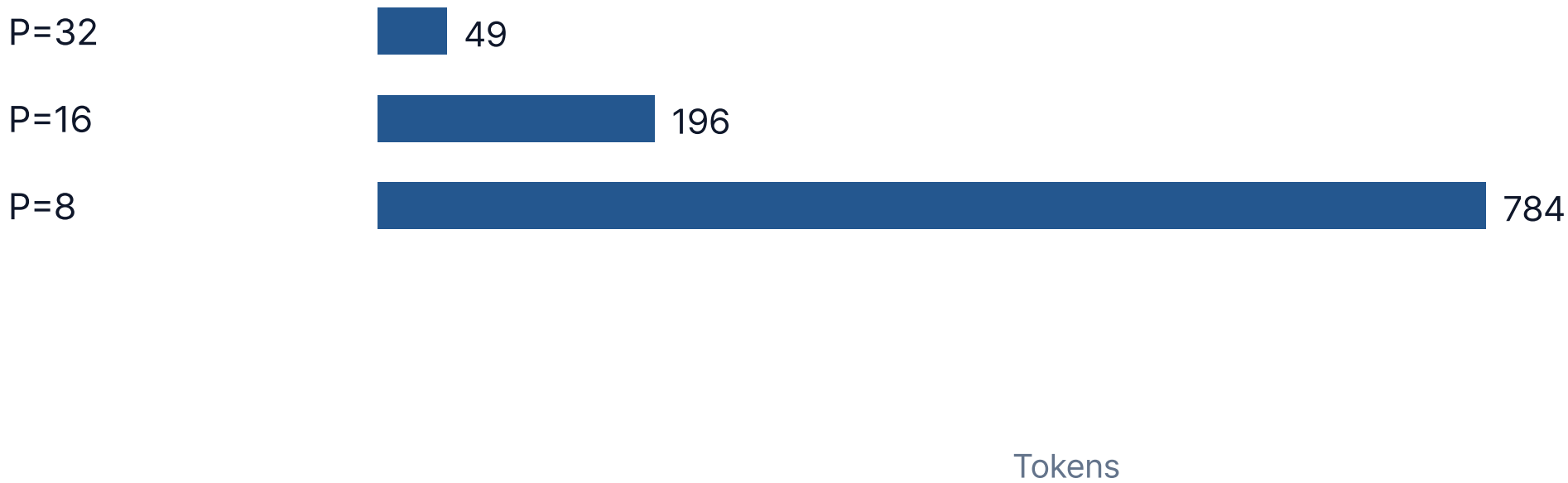
Image patches	p1	p2	p3	p4
Embeddings	z1	z2	z3	z4

Each patch becomes one token

Patch Count

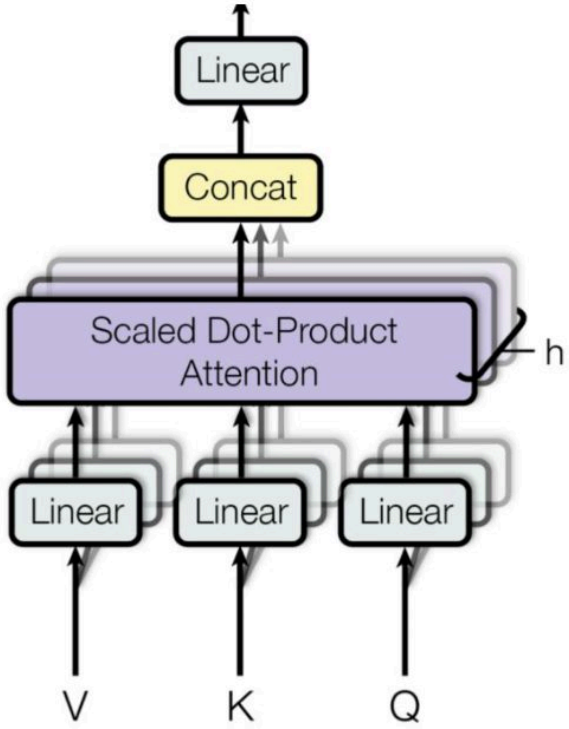
Smaller patches create longer sequences

$$N = (H / P) \times (W / P)$$



Vision Transformer

Patch tokens enter a shared Transformer encoder



Patch tokens share the same encoder block

https://arxiv.org/abs/1706.03762

Position Information

Permutation alone cannot recover the original layout

	Left	Right
Top	top-left	top-right
Bottom	bottom-left	bottom-right

Position identifies where each patch came from

02 · Learning Representations

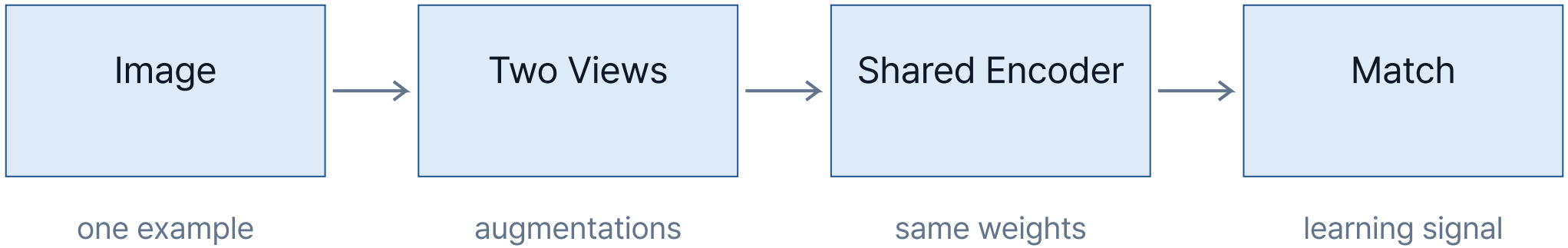
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Why Learn Without Labels

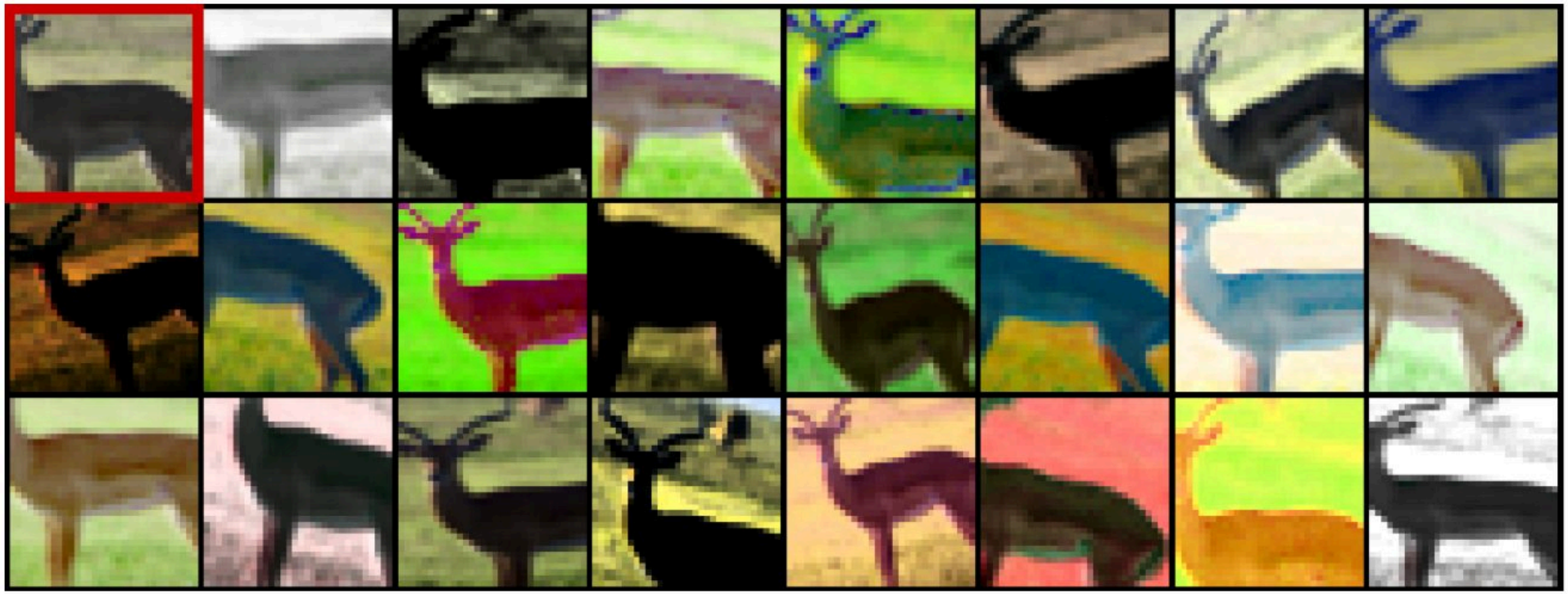
Unlabeled images still provide paired observations



Two views provide a learning relation

Visual Cues

Different crops can preserve the same object



Different views can retain one object identity

https://arxiv.org/abs/1912.01991

Positive and Negative Pairs

A training relation defines which features move together



Pair labels define the contrastive task

Cosine Similarity

Normalization separates direction from vector magnitude

$$\cos(z, v) = z^{T}v \,/\, (||z||\ ||v||)$$

A	1	0
2A	2	0
B	0	1

Similarity Matrix

Aligned views should score above mismatched pairs

	view 1	view 2	view 3
image 1	0.95	0.1	0.2
image 2	0.15	0.9	0.05
image 3	0.1	0.25	0.92

The diagonal marks matched observations

Contrastive Objective

The matching example competes with alternatives

$$L_i = -\log[\exp(s_{ii} / \tau) / \Sigma_j \exp(s_{ij} / \tau)]$$

Representation Collapse

Identical vectors satisfy matching without preserving information

Cat	1	0
Car	1	0
Tree	1	0

Perfect agreement can erase every difference

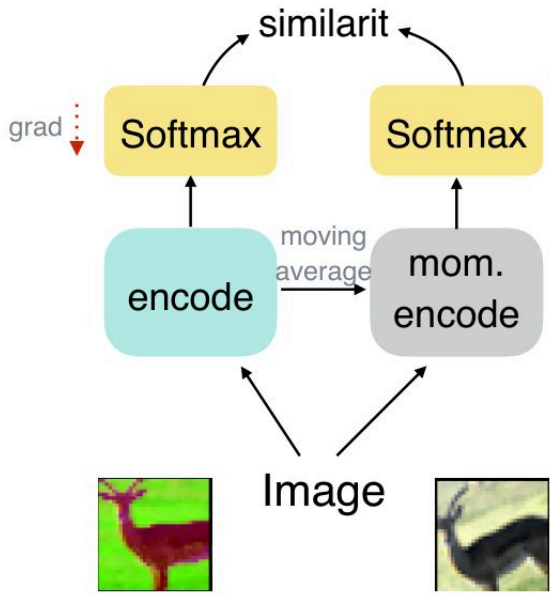
Avoiding Collapse

Additional constraints keep representations informative

Family	Constraint
Contrastive	Compare other examples
Distillation	Teacher targets and asymmetry
Redundancy reduction	Control covariance

Teacher and Student

A teacher supplies targets for another augmented view



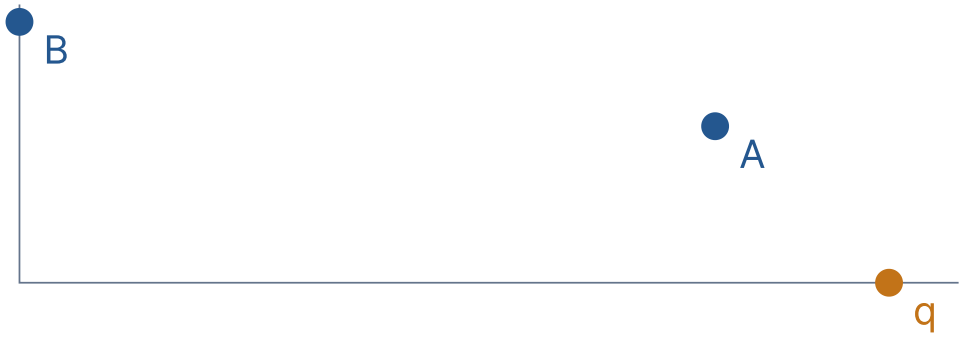
The target comes from another view of the image

https://arxiv.org/abs/2104.14294

Demo: Embedding Distance

Rotate the unit query and compare cosine similarity

Query angle (degrees) 0



q = (1.000, 0.000) | cosine A 0.800, B 0.000

Squared distance A 0.400, B 2.000 | nearest A

Answer: Normalized Neighbors

Nearest neighbors depend on the representation and metric

Query	1	0
Candidate A	0.8	0.6
Candidate B	0	1

Candidate A has the larger cosine similarity

03 · Transferring Features

01 Encoding Images

From pixels to tokens

02 Learning Representations

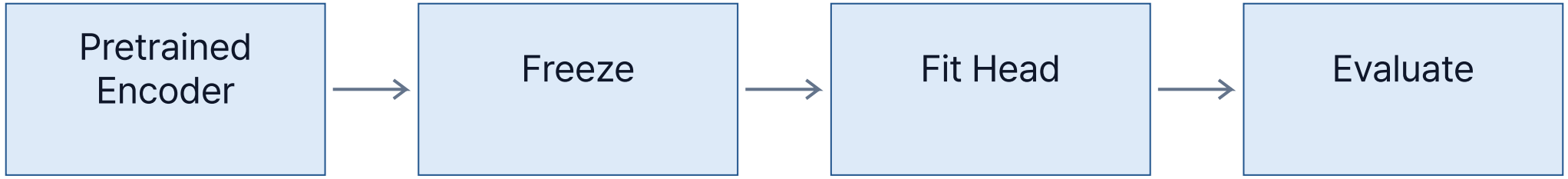
From pairs to features

03 **Transferring Features**

From features to tasks

Frozen Feature Transfer

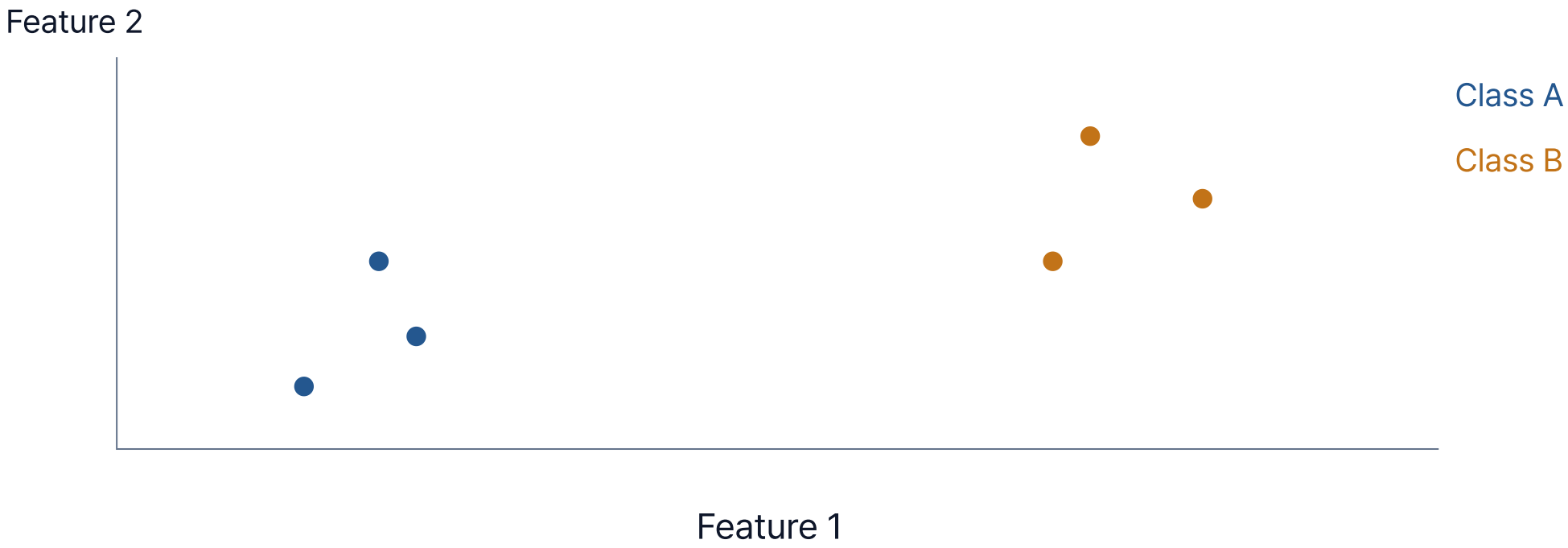
A small readout can reuse a pretrained encoder



Only the readout changes in a frozen probe

Linear Probing

A linear head tests what is already accessible



A simple boundary exposes separable information

Fine-Tuning Scope

Updating more parameters increases adaptation and validation needs



Illustrative trainable parameter units

More trainable weights expand the adaptation space

Augmentation Choice

Preserving the wrong invariant can remove the target

Color	red	green
Meaning	stop	go

Removing color can remove the label

Caveat: Background Shortcut

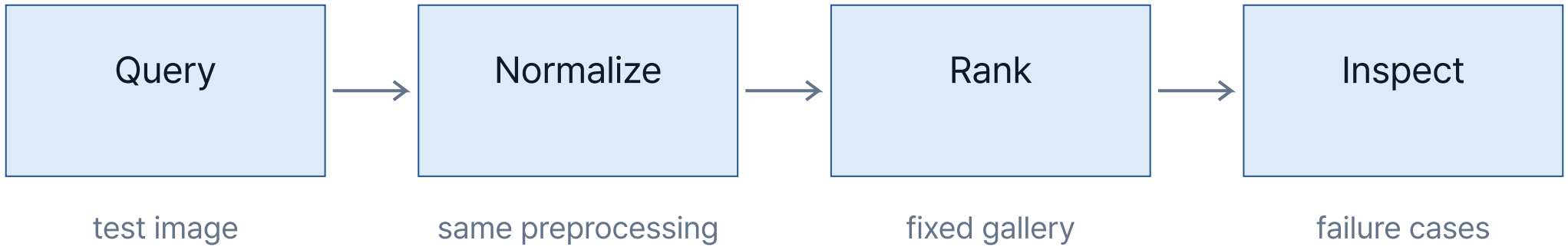
A classifier may learn context instead of the object

	Class A	Class B
Train: snow	8	0
Train: grass	0	8
Test: mixed	4	4

A background shortcut fails when context changes

Retrieval Inspection

Nearest examples reveal useful and spurious similarities



The gallery determines what retrieval can find

Pooling Code

Spatial tokens and global vectors have different shapes

```
tokens = np.arange(24).reshape(2, 3, 4)
global_vectors = tokens.mean(axis=1)
assert global_vectors.shape == (2, 4)
```

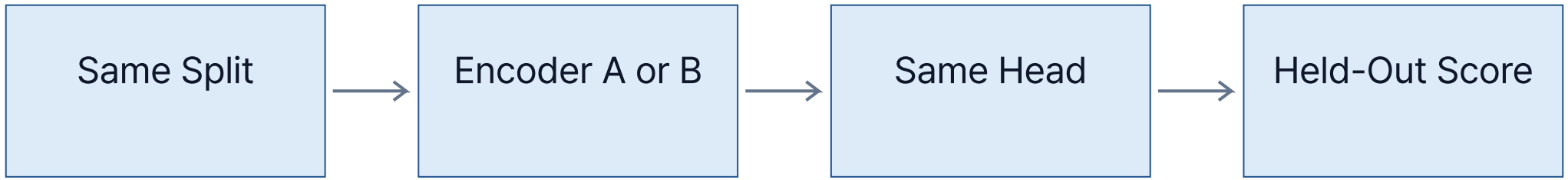
Feature Normalization

Zero-length embeddings need an explicit policy

```
norm = np.linalg.norm(z, axis=-1, keepdims=True)
z_unit = z / np.maximum(norm, 1e-12)
```

Transfer Evaluation

Hold the data split and readout fixed across encoders



A controlled comparison changes one component

Representation Failure Audit

Check semantic, spatial and domain assumptions separately

Question	Check
Meaning	Inspect nearest examples
Location	Keep patch tokens
Domain	Evaluate shifted data

Takeaway: Visual Representations

01 Features determine what a small head can learn

02 Pair construction encodes an invariance assumption

03 Transfer requires a controlled held-out evaluation